

By 洛辞.

Problem 1:

$$\begin{aligned}
 a: \quad \min_{\theta} f(\theta) &= \sum_{i=1}^n w_i (\theta - x_i)^2 \\
 &= \sum_{i=1}^n w_i (\theta^2 - 2x_i\theta + x_i^2) \\
 &= \left(\sum_{i=1}^n w_i\right) \theta^2 - 2\left(\sum_{i=1}^n w_i x_i\right) \theta + \underbrace{\sum_{i=1}^n w_i x_i^2}
 \end{aligned}$$

We see θ as variables.

$$\Rightarrow \theta^* = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (\text{quadratic function!})$$

$$b: \quad f(x) = \max_{s \in [-1,1]} \sum_{i=1}^d s x_i; \quad g(x) = \sum_{i=1}^d \max_{s_i \in [-1,1]} s_i x_i$$

Claim: $f(x) \leq g(x)$, $\forall x$

$$f(x) = \max_{s \in [-1,1]} s \sum_{i=1}^d x_i = \left| \sum_{i=1}^d x_i \right|$$

$$g(x) = \sum_{i=1}^d |x_i| \geq \left| \sum_{i=1}^d x_i \right| = f(x)$$

(Triangle inequality).

$$\begin{aligned}
 c: \quad V &= \frac{1}{6} \cdot 0 + \frac{1}{2} \cdot V + \frac{1}{6} (V+a) + \frac{1}{6} (V-b) \\
 &\Rightarrow V = a - b.
 \end{aligned}$$

$$d: \max_{p^*} L(p^*) \Leftrightarrow \max_{p^*} \log L(p^*)$$

$$\log L(p) = 4 \log p + \log(1-p)$$

$$\log L(p)' = \frac{4}{p} - \frac{1}{1-p} = 0 \Rightarrow p = \frac{4}{5}$$

$$\log L(p)'' = -\frac{4}{p^2} - \frac{1}{(1-p)^2} < 0 \Rightarrow p \text{ maximize ...}$$

Intuitive: Both H/T are independent...

$$\frac{4H}{4H+T} \Rightarrow \frac{4}{5}.$$

$$e: P(A|B) = P(B|A): P(A \cup B) = \frac{1}{3}, P(A \cap B) > 0$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \cap A)}{P(A)} = P(B|A)$$

$$\Rightarrow P(A) = P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{3}$$

$$\Rightarrow 2P(A) > \frac{1}{3} \Rightarrow P(A) > \frac{1}{6} \quad \#.$$

$$f: f(w) = \left(\sum_{i=1}^n \sum_{j=1}^n (\alpha_i^T w - b_j^T w)^2 \right) + \frac{\lambda}{2} \|w\|_2^2,$$

$$\text{which: } w = (w_1, \dots, w_d)^T; \|w\|_2 = \sqrt{\sum_{k=1}^d w_k^2} = \sqrt{w^T w}$$

$$\frac{\partial \frac{\lambda}{2} \|w\|_2^2}{\partial w_i} = 2w_i;$$

$$\frac{\partial (\alpha_i^T w - b_j^T w)^2}{\partial w_k} = 2(\alpha_i^T w - b_j^T w)(\alpha_{ik} - b_{jk});$$

$$\Rightarrow \frac{\partial f(w)}{\partial w} = 2 \sum_{i=1}^n \sum_{j=1}^n (\alpha_i^T w - b_j^T w)(\alpha_i - b_j) + \lambda w. \quad \#.$$

Problem 2:

a: 2 x-axis lines: $C_{n+1}^2 = \frac{n(n+1)}{2}$ ways
2 y-axis lines: $C_{n+1}^2 = \frac{n(n+1)}{2}$ ways
 $\Rightarrow$ One rectangle has: $\frac{n^2(n+1)^2}{4}$ ways
3 rectangles $\rightarrow \frac{n^6(n+1)^6}{4^3}$ ways
 $\Rightarrow O(n^{12})$.

b: $O(n^2)$.

Algorithm 1 Minimum Cost Dynamic Programming

Require: FinalCost = zero array of dimension (n, n) , Cost function $c(i, j)$

FinalCost[0][0] $\leftarrow c(0, 0)$

$\triangleright$ Initialize starting state cost

for $i = 1 : n - 1$ **do**

$\triangleright$ Initialize first column of matrix

FinalCost[i][0] $\leftarrow$ FinalCost[i-1][0] + $c(i, 0)$

end for

for $j = 1 : n - 1$ **do**

$\triangleright$ Initialize first row of matrix

FinalCost[0][j] $\leftarrow$ FinalCost[0][j-1] + $c(0, j)$

end for

for $i = 1 : n - 1$ **do**

$\triangleright$ Calculate remaining values by using dynamic programming

for $j = 1 : n - 1$ **do**

FinalCost[i][j] $\leftarrow$ minimum(FinalCost[i-1][j] , FinalCost[i][j-1]) + $c(i, j)$

end for

end for

FinalCost[n-1][n-1] is the minimum cost

by shauryagoyal
@github.com.

Problem 3 : (Interesting Problem .)